

Recommendation Systems for Personalized Content Discovery

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Problem Understanding:

One of the most significant uses of machine learning is in creating recommendation systems that help people find content to consume (e.g., movies) on platforms like Netflix, Spotify, and YouTube. The biggest hurdle with this type of application is figuring out what to recommend to a given user based on past interactions they had with other users and items, when the bulk of the interactions between users and items are not known.

To address this challenge, a personalized movie recommendation system will be developed using the Netflix Prize Dataset. This system will be capable of learning user preference models, predicting the ratings for new/unseen movies, and generating meaningful Top-K recommendations based on both predicted and given rating scores. As part of the evaluation, we will utilize two metrics: RMSE (root mean square error) and MAP (mean average precision) at 10 (MAP@10).

Dataset Overview:

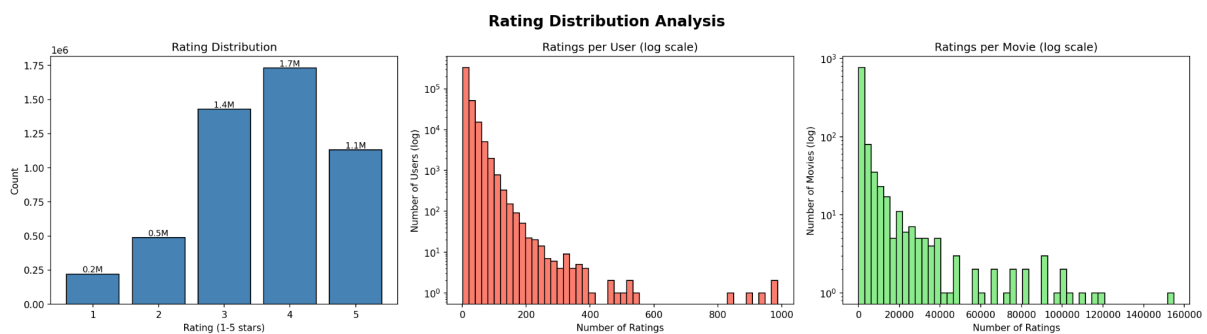
The Netflix Prize Dataset is one of the most heavily researched datasets in the recommendation system field, so it is used extensively as a benchmark in the research community. This project used the first data file (combined_data_1.txt) from the dataset, and had a total of 5,000,000 rating records to ensure that our computations were feasible while still utilizing interesting data patterns.

Property	Value
Ratings loaded	50,00,000
Unique users	4,04,478
Unique movies (in subset)	996
Total movies in catalog	17,770
Rating scale	1 to 5 stars
Date range	December 1999 to December 2005
Matrix sparsity	98.76%

The nature of the data used in this dataset is highly sparse, with a total sparsity of 98.76% (i.e., each user's ratings consist of only a small percentage of the total number of ratings). The issue of sparsity on the user-item interactions is inherent to the real-world recommendation system applications.

Exploratory Data Analysis(EDA):

Rating Distribution:



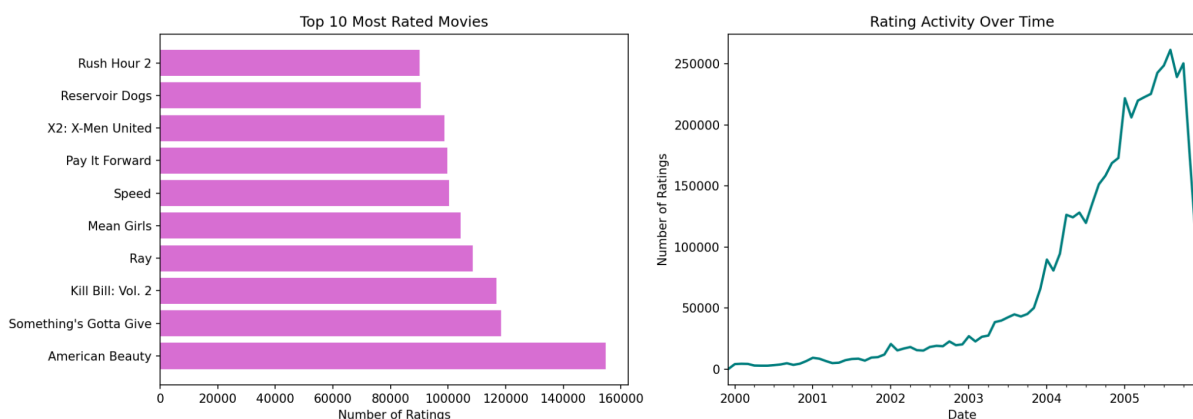
Rating	Count
1 star	220,348 (4.4%)
2 stars	488,238 (9.8%)
3 stars	1,429,692 (28.6%)
4 stars	1,730,291 (34.6%)
5 stars	1,131,431 (22.6%)

The large majority of ratings tend to be positively skewed; more than 57% of the complete ratings are rated at least 4 or 5 stars (on a scale). This indicates that most users will tend to rate a movie that they expect to have a good experience watching, creating an inherent positive bias that needs to be taken into account when developing recommendation models.

User Activity:

The way that users rate films displays a traditional long-tail shape, where there are many users that have only rated a couple of films, while there is a smaller subset of 'super' users that will rate a much larger quantity of films. The relatively small number of 'super' users creates a cold-start problem for those infrequent users because they rely on the recommendations of the 'super' users to find good films.

Movie Popularity:



The film ratings plot also demonstrates a power-law distribution in that there are very few films that receive a majority of the ratings, while there are many films that receive very few ratings. This creates a popularity bias, which can lead a naive model to continually suggest the same popular films repeatedly.

Sparsity:

The user-movie interactions that form the user-movie interaction matrix are highly sparse, with only 1.24% of all possible user-movie pairs having an observed rating, resulting in an 98.76% level of sparsity in the matrix. Therefore, direct memory-based approaches are not possible and a Latent Factor Model would be required for the user-movie recommendation engine to deal with sparse data and to address the issues of popularity bias and providing recommendations to new users and new content.

Methodology:

Data Preprocessing:

The raw Netflix data files are in a nonstandard format, wherein each movie ID is listed on its own line with a trailing colon, followed by rating records. A custom user-defined parser was created to accommodate this format. Ratings were filtered for users with at least 20 ratings and movies with a minimum of 10 ratings to maintain sufficient signal for training the model. An 80/20 train-test split was performed, dividing the dataset into a training set and a held-out test set to allow for unbiased testing.

Model Selection Rationale:

Four different models are executed, ranging from classical methods to those based on neural approaches:

SVD (Singular Value Decomposition): Decomposes the user-item ratings matrix into latent user and item factor vectors. Each user and item is depicted in a low-dimensional space, whereas predictions of the ratings are made from the dot products of those vectors and bias terms. Trained by a stochastic gradient descent with regularization.

NMF (Non-Negative Matrix Factorization): Similar to SVD, but imposes non-negativity constraints over all factor matrices, thus leading to more interpretable latent factors, which can be loosely associated with genre or thematic clusters.

SVD++ (Extended SVD with Implicit Feedback) is an enhanced version of the standard SVD that also takes into account implicit feedback (the movies a user has rated regardless of the rating value) to reflect the user's interest in a movie the act of rating represents.

NCF (Neural Collaborative Filtering) is a deep-learning method that replaces the dot product used in SVD with a multi-layered neural network. To generate the input for the neural network, the User ID and Movie ID are mapped to 64-dimensional embedding vectors which are concatenated and then passed through fully connected (dense) layers with relu activations and dropout regularization. The model will be trained using the Adam optimizer with mean square error loss over a sample size of 500,000.

Model Design:

Hyperparameter	SVD	NMF	SVD++	NCF
Latent factors	100	15	20	64
Epochs	20	20	20	5
Learning rate	0.005	default	default	0.001
Regularization	0.02	default	default	Dropout 0.2
Training samples	1,000,000	200,000	200,000	500,000
Optimizer	SGD	SGD	SGD	Adam

Train-Test Split Methodology:

- To split the filtered dataset into an 80% training dataset and a 20% testing dataset, the nature of random sampling was used while a fixed random seed value (42) was employed to ensure the sampling was reproducible.
- The split has been done at the rating level and not the user level so that each dataset contains ratings from the same users.
- All models will be trained only with data from the training data prior to evaluating on test data.

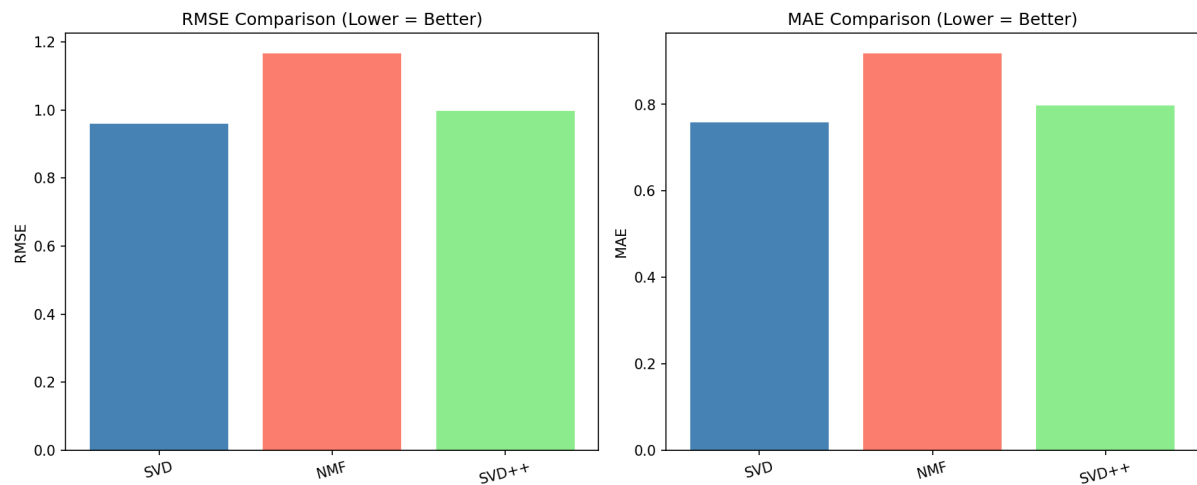
Evaluation Metrics:

The accuracy of a prediction can be measured using RMSE or root mean square error, which indicates how closely predicted ratings are to actual ratings. A low rate of accuracy will be treated as a preferred measure of similarity.

If the rank at which relevant movies are found near the top of the Top-10 movie recommendations is determined, MAP@10 or Mean Average Precision at 10 will provide an accurate indication of the degree of rank at which relevant movies are available to the user. A rating of 3.5 or higher was determined to be a relevant rating for a given movie.

A ranking of movies to users will be calculated by determining the predicted rating of all 10 top-ranked movies to each user. The precision will be calculated as the total number of relevant movies divided by the ranking of each relevant movie in the Top 10. The average precision will equal the mean precision across individual users, while MAP@10 will equal the mean across all users.

Results:



Model	RMSE	MAP@10	MAE
SVD	0.9594	0.6933	0.7580
NMF	1.1669	0.5819	0.9184
SVD++	0.9966	0.5862	0.7980
NCF	0.9941	N/A	-

The SVD model provides the best RMSE (0.9594) and the best MAP@10 (0.6933) performance. Therefore, the SVD model is the optimal choice for making recommendations. Despite an RMSE of 0.9941 and fewer training examples, the performance of the NCF model is still competitive.

Recommendation Generation & Quality:

Sample Recommendations:

Top 10 Recommendations for User 2636813:

1. Sex and the City: Season 4 (2001) – Predicted: 4.62 STAR
2. Futurama: Monster Robot Maniac Fun Collection (1999) – Predicted: 4.61 STAR
3. Christmas with The Simpsons (1989) – Predicted: 4.51 STAR
4. Lord of the Rings: The Return of the King: Extended Edition: Bonus Material (2003) – Predicted: 4.42 STAR
5. Chappelle's Show: Season 1 (2003) – Predicted: 4.42 STAR
6. Bill Cosby: Himself (1983) – Predicted: 4.42 STAR

7. Star Trek: The Next Generation: Season 7 (1993) – Predicted: 4.38 STAR
8. Les Miserables in Concert (1996) – Predicted: 4.35 STAR
9. The Twilight Zone: Vol. 36 (1961) – Predicted: 4.34 STAR
10. Cadfael: A Morbid Taste for Bones (1996) – Predicted: 4.32 STAR

Top 10 Recommendations for User 530789:

1. Yojimbo (1961) – Predicted: 4.96 STAR
2. Ray (2004) – Predicted: 4.85 STAR
3. The Deer Hunter (1978) – Predicted: 4.83 STAR
4. Ghosts of Rwanda: Frontline (2004) – Predicted: 4.80 STAR
5. Sense and Sensibility (1981) – Predicted: 4.78 STAR
6. Futurama: Monster Robot Maniac Fun Collection (1999) – Predicted: 4.77 STAR
7. Lord of the Rings: The Return of the King: Extended Edition: Bonus Material (2003) – Predicted: 4.76 STAR
8. Seven Samurai (1954) – Predicted: 4.73 STAR
9. Xena: Warrior Princess: Season 3 (1995) – Predicted: 4.72 STAR
10. The Twilight Zone: Vol. 12 (1962) – Predicted: 4.68 STAR

Top 10 Recommendations for User 901650:

1. Les Miserables in Concert (1996) – Predicted: 4.68 STAR
2. The Lady Eve (1941) – Predicted: 4.66 STAR
3. Star Trek: The Next Generation: Season 7 (1993) – Predicted: 4.60 STAR
4. Chappelle's Show: Season 1 (2003) – Predicted: 4.52 STAR
5. The Three Stooges: Three Smart Saps (1942) – Predicted: 4.50 STAR
6. Futurama: Monster Robot Maniac Fun Collection (1999) – Predicted: 4.49 STAR
7. The Twilight Zone: Vol. 36 (1961) – Predicted: 4.49 STAR
8. Foyle's War: Set 2 (2003) – Predicted: 4.48 STAR
9. The Chorus (2004) – Predicted: 4.48 STAR
10. High and Low (1963) – Predicted: 4.45 STAR

Success Cases:

The following users received 100% of their Top-10 recommended movies as relevant (i.e., they received an actual rating of ≥ 3.5): 901650, 767537, 2641990, 1252974, and 2234150.

Because these five users all rated a large number of movies, the SVD model was able to predict their preferred movie ratings very accurately and target their movie ratings via latent factor captures accurately.

Failure Cases:

The following table lists the User IDs of those whose Top 10 Recommendation's were at least Zero Percent Relevant for Users Identified in the prior (1) Section (0 Percent Relevant):

User ID - 17462 (38 training ratings); User ID - 880344 (15 ratings); User ID - 163069 (20 ratings); User ID - 1825371 (55 ratings); User ID - 2239630 (17 ratings);

Key observation: Sparse rating history is to blame for nearly all failure cases above (15 to 55 User Training Ratings). The model is unable to accurately estimate for those Users' latent preference vectors due to a lack of data, this classic cold-start problem creates inadequacies for the model.

Key Insights:

- Overall, SVD shows significant performance advantages over NMF and SVD++ in both RMSE and MAP@10. This shows that classic matrix factorization with bias terms is a robust baseline for this dataset.
- NMF had the highest RMSE (1.1669) among the implemented algorithms due to its non-negativity constraints limiting expressiveness even though its factors were more interpretable.
- SVD++ did show marginal improvement (over NMF) in the recommendation quality through utilizing implicit feedback albeit SVD++ was limited in its ability to learn implicit feedback patterns because of the 200k training data set.
- The NCF model achieves a training RMSE of 0.9941 with only 5 epochs of training completing as it shows a steady tendency to improve in RMSE by epoch (1.1065 to 0.9941) and improvements continue with additional epochs. Therefore it can be reasonably expected that further training would yield even greater improvements.
- Due to the very high sparsity (98.76%) of the dataset this problem is highly suitable for latent factor models as opposed to memory-based recommenders.
- Using the MAP@10 of 0.6933, SVD recommended 7 out of every 10 received Recommended Titles to be highly rated by Users. Very good user's enjoyability for the Recommended Movies.

Future Improvements:

- Cold Start Handling: Use content-related aspects of users/movies (i.e., genre, release date) for users/movies that haven't been rated.
- Full Dataset Training: Using all combined_data in 4 datasets (100M ratings) will grow the accuracy of a model's predictions.
- Hybrid System: Combine SVD with metadata using content-based filtering for more accurate predictions.

- Training of extended NCF : Training the extended NCF with a GPU for >20 epochs will likely result in an RMSE <0.95.
- Temporal Aspects: Because users' preferences change over time, using the time stamp of a rating will help improve personalization.
- Deploying the SVD model in a FastAPI service would allow real-time recommendations to be generated.

References:

1. Netflix Prize Dataset: <https://www.kaggle.com/datasets/netflix-inc/netflix-prize-data>
2. Koren, Y. (2008). Factorization meets the neighborhood: a multifaceted collaborative filtering model. KDD '08
3. He, X. et al. (2017). Neural Collaborative Filtering. WWW '17
4. Scikit-Surprise library: <https://surpriselib.com>